Financial Management

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5.1 & 5.2: Capital Structure Theories

Relationship between Financial Leverage and Cost of Capital

- *If leverage affects the cost of capital and the value of the firm, an optimum capital structure would be obtained at that combination of debt and equity that maximises the total value of the firm or minimises the weighted average cost of capital .*
- *The existence of an optimum capital structure is not accepted by all. David Durand identified two extreme views*

1. ***Net Income Approach***
2. ***Net Operating Income Approach***

Relationship between Financial Leverage and Cost of Capital

Net Income Approach	Net Operating Income Approach	Traditional Approach
The cost of debt and cost of equity are assumed to be independent to the capital structure. The WACC and the total value of the firm rises with increased use of leverage	The cost of equity is assumed to be increase linearly with leverage. As a result, the WACC remains constant and the total value of the firm also remains constant as leverage is changed.	The cost of capital declines and the value of the firm increases with leverage up to a prudent debt level and after reaching the optimum point (minimum cost of capital or maximum value of the firm), coverage causes the cost of capital to increase and the value of the firm to decline.

Assumptions and Definitions

To examine the relationship between financial leverage and cost of capital the following simplifying assumptions are commonly made:

- Firms employ only two types of capital: debt and equity
- The total assets of the firm are given. The degree of leverage can be changed by selling debt to repurchase shares or selling shares to retire debt
- Investors have identical subjective probability distributions of net operating income for each company
- The net operating income is not expected to grow or decline over time
- The firm pursues a policy of paying all of its earnings as dividends. (100 % dividend pay out ratio)
- The business risk is assumed to be constant and independent of capital structure and financial risk.
- There is no income tax, corporate or personal.

Assumptions and Definitions

Based on the discussed assumptions, the analysis focuses on the following rates:

$k_d = \frac{INT}{B} = \frac{Annual\ Interest\ Charges}{Market\ Value\ of\ Debt}$; So B = $\frac{INT}{k_d}$

$k_e = \frac{D_1}{P_0}$ + g ;

If Dividend = EPS then g = 0

Therefore, $k_e = \frac{D_1}{P_0} + 0 = \frac{EPS_1}{P_0}$

Lets denote P_0 = S = Market value of the equity.

Assumptions and Definitions

When the dividend pay out ratio is 100 % , and earnings constant, k_e, as defined here, represents the cost of equity capital

$$k_o = \frac{O}{V} = \frac{Net\ Operating\ Income}{Market\ Value\ of\ the\ Firm}$$

where

V = B + S

k_o = over-all capitalization rate of the firm.

Since it is the weighted average cost of capital,

$$k_o = k_d\left(\frac{B}{B+S}\right) + k_e\left(\frac{S}{B+S}\right)$$

Net Income Approach (NI)

- According to this approach, the cost of debt capital, k_d, and the cost of equity capital, k_e, remain unchanged when B/S, the degree of leverage, varies. The constancy of k_d and k_e with respect to B/S means that k_o, the average cost of capital is measured as

$$k_o = k_d\left(\frac{B}{B+S}\right) + k_e\left(\frac{S}{B+S}\right)$$

- Declines as B/S increases. This happens because when B/S increases, k_d, which is lower than k_e, receives a higher weight in the calculation of k_o.

NI Approach: Illustration

- There are two firms A and B similar in all respects except in the degree of leveraged by them. Financial data for these firms are shown below. Calculate k_o . Amount in (Rs.)

Particulars	Firm A	Firm B
Net Operating income (EBIT)	10000	10000
Interest on debt	0	3000
Equity earnings	10000	7000
Cost of equity capital	10%	10%
Cost of debt capital	6%	6%
Market value of equity	100000	70000
Market value of debt	0	50000
Total value of the firm	100000	120000

Net Operating Income Approach (NOI)

- According to the NOI approach, the overall capitalisation rate and the cost of debt remain constant for all degrees of leverage.

$$k_e = k_o + (k_o - k_d)\left(\frac{B}{S}\right)$$

- The critical premise of this approach is that the market capitalises the firm as a whole at a discount rate which is independent of the firm's degree of leverage.
- As a consequence, the division between debt and equity is irrelevant.
- An increase in the use of debt funds which are 'apparently cheaper' is offset by an increase in the equity capitalisation rate.
- This happens because equity investors seek higher compensation as they are exposed to greater risk arising from increase in the degree of leverage.
- They raise the capitalisation rate as the degree of leverage increases

NOI Approach

- The market value of a firm depends on its net operating income and business risk.
- The change in degree of leverage employed by a firm cannot change these underlying factors.
- It merely changes the distribution on income and risk between debt and equity without affecting the total income and risk which influence the market value of the firm.
- Hence, the degree of leverage per se cannot influence the market value of the firm.

NOI Approach: Illustration

Two firms A and B, are similar in all respects except the degree of leverage employed by them. Relevant financial data for these firms are given below. Calculate k_e.

Particulars	Firm A	Firm B
Net Operating income (EBIT)	10000	10000
Overall capitalization rate (Ko)	0.15	0.15
Total market value (V)	66667	66667
Interest on debt (INT)	1000	3000
Debt capitalisation rate (Kd)	0.10	0.10
Market value of debt (B)	10000	10000
Market value of equity (S)	56667	36667
Degree of leverage	0.176	0.818

Modigliani and Miller Position : Assumptions

- Restated and amplified the NOI position in terms of three basic propositions
- Capital markets are perfect. Information is freely available and transactions are costless; securities are infinitely divisible.
- Investors are rational. Investors are well-informed and choose a combination of risk and return that is most advantageous to them.
- Investors have homogeneous expectations. Investors hold identical subjective probability distribution about future operating earnings.
- Firms can be grouped into ‘equivalent risk classes’ on the basis of their business risk.
- There is no corporate tax.

MM's Basic Propositions

Based on the assumptions, MM derived the following three propositions.

1. ***Proposition I :*** The total market value of a firm is equal to its expected operating income divided by the discounted rate appropriate to its risk class. It is independent of the degree of leverage.

$$V_j = S_j + B_j = O_j / k_o$$

Where,

V_j = total market value of firm *j*

S_j = market value of the equity of firm *j*

B_j = market value of the debt of firm *j*

O_j = expected operating income of firm *j*

k_o = discount rate applicable to the risk class to which the firm *j* belongs.

MM's Basic Propositions

2. ***Proposition II :*** The expected yield on equity is equal to k_o plus a premium. This premium is equal to the debt-equity ratio times the difference between k_o and the yield on debt (k_d).

$$k_e = k_o + (k_o - k_d)\left(\frac{B}{S}\right)$$

3. ***Proposition III :*** The cutoff rate for investment decision making for a firm in a given risk class is not affected by the manner in which the investment is financed. Investment and financing decisions are independent because the average cost of capital is not affected by the financing decision.

Arbitrage Mechanism

- According to MM, if two firms, X and Y, are in the same risk class and have the same expected operating income, they will have the same value in the market place, irrespective of differences in the capital structure.
- If their values diverge, investors will restore to arbitraging.
- Investors will sell the securities of the firm that has a higher value and buy the securities of the firm that has lower value.
- The arbitrage actions of investors will bring about an equality in the value of the two firms.

MM Approach: Illustration

Two firms X and Y, similar in all respects except in their capital structure. Firm X is financed by equity only; firm Y is financed by a mixture of equity and debt. Relevant financial data of the two firms are given below:

Particulars	X	Y
Total Capital Employed	1000000	1000000
Equity capital	1000000	600000
Debt	0	400000
Net Operating Income	100000	100000
Debt Interest	0	20000
Market Value of Debt (Debt capitalization rate is 5%)	0	400000

MM Approach: Illustration

Particulars	X	Y
Equity earnings	100000	80000
Equity capitalization rate	10%	12%
Market value of equity	1000000	666667
Total market value of the firm	1000000	1066667
Average cost of capital	10%	9.37%
Debt-Equity ratio (in terms of market value)	0	0.6

Show Arbitration according to MM approach

MM Approach: Illustration

Equity investors would do well to sell their equity investment in firm Y and invest in the equity of firm X with personal leverage.

An equity investor who owns 1% equity in firm Y would do well to:

- Sell his equity in firm Y for Rs.6667
- Borrow Rs. 4000 at 5 % interest on personal account and
- Buy 1.0667 per cent of the equity of firm X with the amount of Rs. 10667 that he has.

Such an action will result in the following income:

Income on investment in firm X	= Rs. 1066.7
less interest (Rs.4000 * 0.05)	= Rs. 200
Net Income	= Rs. 866.7

MM Approach: Illustration

- This net income of Rs. 866.7 is higher than a net income of Rs. 800 forgone by selling 1% of equity of firm Y and the leverage ratio is the same in both the cases.
- When investors sell their equity in firm Y and buy the equity in firm X with personal leverage, the market value of equity of firm Y tends to decline and the market value of equity of firm X tends to rise.
- This process continues unlit the net market values of both the firms become equal because only then the possibility of earning a higher income for a given level of investment and leverage by arbitraging is eliminated.
- As a result, the cost of capital for both the firms is the same

Setting a target Capital Structure : EBIT-EPS Analysis

Helps in choosing an optimum capital structure

Analyse how sensitive earnings per share (EPS) is to changes in earnings before interest and tax (EBIT) under different financing alternatives.

$$EPS = \frac{(EBIT - I)(1-t)}{n}$$

Where,

EPS = earning per share

EBIT = earning before interest and tax

I = interest burden

t = tax rate

n = number of equity shares

EBIT-EPS: Illustration

Falcon Ltd plans to raise additional capital of Rs. 10 million for financing an expansion project. In this context, it is evaluating two alternative financing plans.

(i) Issue of equity shares worth 1 million at Rs. 10 per share, and

(ii) Issue of debenture carrying 14 per cent interest.

What will be the EPS under the two alternative financing plans for two level of EBIT say Rs. 4 million and Rs. 2 million?

Tax rate is 50%. Existing capital structure includes 1million equity shares of Rs. 10 each.

Explaining Financial Choices

Two theories are commonly advanced to explain real world corporate financing behaviour, viz. the trade-off theory and pecking order theory.

1. Trade-off Theory:

- Trade-off between the tax shelter provided by debt and the cost of financial distress.
- According to this theory, profitable firms with stable, tangible assets would have higher debt-equity ratios. On the other hand, unprofitable firm with risky, intangible assets tend to have lower debt-equity ratio.

Explaining Financial Choices

2. Pecking Order Theory

- Explains why profitable firms use little debt.
- According to this theory, there is a pecking order of financing which goes as follows:

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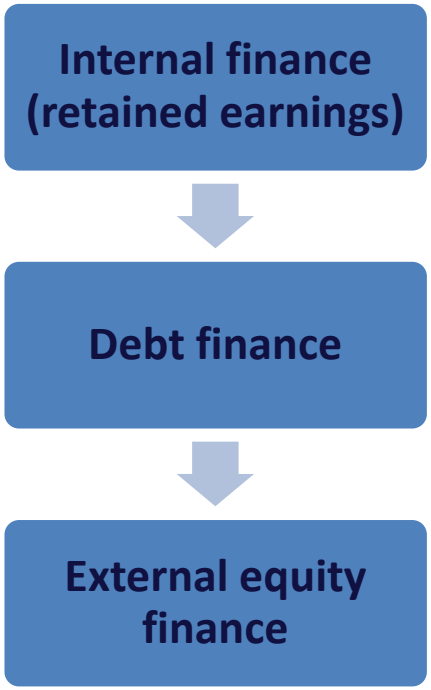

Determinants of Capital Structure Decision

1. Asset Structure
2. Stability of revenues
3. Operating leverage
4. Growth rate
5. Profitability
6. Taxes
7. Control
8. Attitude of management
9. Attitude of lenders and rating agencies

Practice Problems

1. Calculate EPS of Solid Ltd. and Sound Ltd. Assuming (a) 20% before tax rate of return on assets, (b) 10% before-tax rate of return on assets based on the following data.

Particulars	Solid Ltd. (Rs. In Lakh)	Sound Ltd. (Rs.in Lakh)
Assets	100	100
Debt	---	50 (12% debenture & loan)
Equity	100 (Rs. 10/share)	50 (Rs. 10/share)

Assume a 50% tax rate in both the cases. Give your comments on the financial leverage.

Practice Problems

2. M.C. Ltd. Is planning an expansion programme which will required Rs. 30 crores and can be funded through one of the three following options
   (a) Issue further equity share of Rs. 100 each at par.
   (b) Raise loans at 15% interest
   (c) Issue preference share at 12%

   Present paid up capital is Rs. 60 crores and average annual EBIT is Rs. 12 crores. Assume IT rate at 50%. After the expansion, EBIT is expected to be Rs. 15 crores per annum.

   Calculate EPS under the three financing options indicating the alternative giving the highest return to the equity shareholders.

THANK YOU!!!